GRACEDAVE J. TORSINO

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PROFESSIONAL SUMMARY

Computer Science graduate (Cum Laude) with experience in **software development, robotics, artificial intelligence**, and **experimental research**. Experienced in developing software solutions, data collection and evaluation pipelines, conducting experimental research, and solving technical problems through programming and systematic analysis.

SKILLS

Programming / Database: C, Python, JavaScript, TypeScript, PostgreSQL, SQL
Development Tools: Git, GitHub, Conda
Machine Learning & Computer Vision: PyTorch, Diffusion Policy, OpenCV, YOLOv5
Robotics & Simulation: ROS2, Webots, Teleoperation Systems
Operating Systems / Productivity: Windows, Linux (Ubuntu), Google Workspace, Microsoft Office

EXPERIENCE

AI and Robotics Laboratory (AIRLab) at Caraga State University – Main Campus — *Intern*
Project: Teaching RoArm-M2 Push and Pick-and-Place Behaviors
March 2026 – May 2026

- Adapted a vision-based Diffusion Policy and developed a leader–follower teleoperation workflow for push and pick-and-place tasks using RoArm-M2 robotic arms
- Implemented sim-to-real transfer experimentation and policy evaluation

PROJECTS

6d CIMS — Customizable Inventory Management System

- Developed a customizable inventory management system that allows organizations to define their own inventory tables and fields
- Implemented tool borrowing and partial-return tracking, consumed-material records with supporting attachments, and low-stock and overdue alerts
- Designed for offline use as a web application or Android app, with backup and restore functionality

Prototype 0 (<p.0>): Custom Imperative Programming Language

- Designed and implemented a custom imperative programming language and its C-based compiler
- Designed the application to run offline as a web app or Android app, with backup and restore functionality

StateFlow: Automata Simulator

- Developed a web-based application that converts natural-language descriptions into finite automata, visualizes state transitions, and simulates user-provided input strings through an AI-powered backend

Individual Undergraduate Thesis: Sim-to-Sim Transfer of Low-Dimensional Diffusion Policy for SO-ARM100 Pick-and-Place Using Webots Demonstrations

- Conducted training and evaluation of a low-dimensional Diffusion Policy for SO-ARM100 pick-and-place tasks in Webots using demonstrations collected through real-to-sim teleoperation
- Developed teleoperation-based data collection and evaluation pipelines for sim-based Diffusion Policy experiments
- Adapted and integrated an existing Diffusion Policy implementation into a custom Webots workflow via dataset, task, and workspace configuration
- Designed and implemented environment randomization for data collection, policy evaluation, and generalization testing in Webots

Real-Time Face Anonymization (Python + OpenCV)

- Developed a real-time face anonymization system using OpenCV for privacy-preserving video processing

3D Projectile Simulator

- Developed a physics-based 3D projectile motion simulator with interactive visualization

EDUCATION

Caraga State University – Main Campus — *BS in Computer Science (2022 – 2026)*
Honors: Cum Laude

PRESENTATION

Students' Congress on Innovation, Informatics, and Computing (SCII-Com) 2026
Research Presenter — *May 22, 2026*